Software Engineering - CS266

Group 3

Fourth Semester Project

Code - Playground-Coding-Platform

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Abstract

Code Playground is a comprehensive web-based coding platform developed to enhance programming skills through practice and competition. The system provides an interactive environment where users can solve problems, participate in coding contests, track their progress, and engage with a community of learners. The platform supports multiple programming languages, real-time code execution, and progress tracking features.

Key features include:

- Problem categorization by difficulty.
- Coding contests with leaderboard.
- Personalized progress tracking.
- Community discussion forums.
- Secure authentication.

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1 Project Objectives

Code Playground - a powerful yet easy-to-use platform where programmers can improve their skills by solving coding problems and competing in contests. Here's what we aim to achieve:

Practice Coding

- Solve problems sorted by difficulty (easy/medium/hard).
- Work with different programming languages (C++, Java, Python).

Join Coding Contests

- Participate in timed competitions
- Create custom contests with chosen problems

Track Progress

- See personal stats and improvement over time.

Learn Together

- Discuss solutions with other users
- Share tips and techniques

Easy to Use

- Clean, simple design that works on computers
- Helpful features to make coding easier

2 System Features

User Roles

1. Admin: Full system access including problem management, contest moderation, and user management
2. Regular User: Can solve problems, participate in contests, track progress, and interact with community
3. Contest Host: Users who can create and manage coding contests

Core Functionalities

- Problem Solving Interface
- Multiple language support
- Test case run
- Submission history

Contest System

- Real-time leaderboard
- Time-bound challenges
- Problem scoring based on solved problems
- Contest leaderboard

Progress Tracking

- Solved problems statistics
- Skill assessment
- Solved problem history

Community Features

- Solution discussions
- feedback
- User rankings
- Collaborative learning

3 System Architecture

Presentation Layer

- Dynamic web interface using EJS templates
- Styled with CSS for responsive design
- Monaco Editor (used in VS Code) for code editing
- HTML5 for structure and accessibility
- Client-side JavaScript for interactive features

Application Layer

- Node.js with Express.js framework
- Key functional modules:
    - Code execution service
    - Contest management
    - system User authentication handlers
    - Problem repository interface
- JSON Web Tokens (JWT) for secure sessions

Data Layer

- MongoDB database for storing:
    - User profiles and credentials
    - Problem sets and metadata
    - Contest details and submissions

Supporting Services

- JWT-based authentication system
- Monaco Editor integration for advanced code editing
- MongoDB Atlas for cloud database management
- Local file system for efficient code storage

4 Technology Stack

Frontend

- EJS (Embedded JavaScript) for server-side templating
- CSS3 for styling and responsive design
- HTML5 for page structure and semantics
- Monaco Editor for code features
- JavaScript for client-side interactivity

Backend

- Node.js with Express.js framework
- MongoDB with native driver for database operations
- JSON Web Tokens (JWT) for authentication
- Express-Session for session management

Development & Deployment

- Git for version control
- MongoDB Compass for database management Visual Studio Code as primary IDE

Hosting Environment

- MongoDB Atlas for cloud database service
- Local file storage for code submissions and test cases

5 Data Management

Database Schema

- Users:Stores user credentials.
- Problems:Contains problem statements, test cases.
- Submissions:Records all code submissions.
- Contests:Manages contest information and participation
- Discussions: Stores community interactions

Data Flow

- User submits code through web interface
- System stores submission in database
- Code is sent to execution engine
- Results are processed and stored
- Analytics are updated in real-time

6 Technical Implementation Details

6.1 Authentication System

The authentication system is implemented using JWT (JSON Web Tokens):

- Passwords are hashed using bcrypt for security
- Token-based sessions manage user state
- Profile pictures are handled via file uploads

6.2 Code Execution Engine

The code execution system:

- Supports multiple programming languages
- Executes code in a sandboxed environment
- Validates code against test cases
- Provides detailed execution results and error messages

6.3 Contest Management

The contest system features:

- Time-based contest status updates (Upcoming, Ongoing, Completed)
- Automatic leaderboard management
- Contest registration and participation handling
- Rating system for contestants

6.4 Problem Management

Problems are managed with:

- Difficulty categorization (Easy, Medium, Hard)
- Test case definition and validation
- Problem searching and filtering
- Solution submission tracking

7. User Interface

7.1 Home Page

- Introduction to the platform
- Quick access to key features
- Platform benefits and testimonials

7.2 Problem Page

- List of available problems with filtering options
- Problem cards showing difficulty and completion status
- Search functionality for finding specific problems

7.3 Code Editor

- Split-pane interface with problem description and code editor
- Language selection dropdown
- Code execution, testing, and submission buttons
- Result display with test case details

7.4 Contest Page

- Current and upcoming contest listings
- Contest details including start time, duration, and problems
- Registration options for contests
- Leaderboard for ongoing and completed contests

7.5 Profile Page

- User statistics and achievements
- Solved problems and participation history
- Submission records with language and status
- Profile management options

8. System Workflows

8.1 Problem Solving Workflow

- User browses or searches for problems
- User selects a problem to solve
- User writes code in the editor
- User tests code against example test cases
- User submits solution for evaluation
- System validates solution against all test cases
- User receives feedback on solution correctness

8.2 Contest Participation Workflow

- User browses available contests
- User registers for a contest
- During contest window, user accesses contest problems
- User solves problems and submits solutions
- System updates leaderboard based on submissions
- Contest ends and final rankings are calculated
- User ratings are updated based on performance

8.3 Discussion Workflow

- User browses or creates discussion topics
- Users can post comments on discussions
- Community members can upvote or downvote content
- Discussions can be tagged for categorization
- Users can search for specific discussion topics

9. Security Considerations

9.1 Authentication

- Password hashing with bcrypt
- JWT token-based authentication
- Session management and expiration

9.2 Code Execution

- Sandboxed execution environment
- Input validation to prevent injection attacks
- Resource limiting to prevent DoS attacks

9.3 Data Privacy

- User data protection
- Restricted access to private contests
- Role-based permissions (admin vs. regular users)

10 Future Scope

Enhanced Features

- AI-assisted problem recommendations
- Collaborative coding environment
- Video tutorial integration
- Mobile application version

Technical Improvements

- Support for additional languages
- Distributed code execution
- Advanced analytics dashboard
- Offline capabilities

Community Expansion

- Mentor-mentee programs
- Coding bootcamps
- University partnerships
- Job placement integrations

Research Opportunities

- Learning pattern analysis
- Problem difficulty calibration
- Code quality assessment
- Plagiarism detection

11. Conclusion

Code Playground provides a comprehensive solution for coding practice, competitive programming, and community engagement. With its robust architecture, intuitive interface, and feature-rich environment, it offers an excellent platform for programmers to enhance their skills, participate in contests, and engage with the programming community.

The project demonstrates good software engineering practices in terms of architecture, code organization, and feature implementation, making it maintainable and extensible for future development.